字符串入门

人员

臧晨云 隋溟琛 霍佳铭 于昊华 刘子郴 何佩瑜 刘克阳 初星熠 陈之恒 刘奕男 李景翾 张高 到课

上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/4029

王向东老师周六一点半C++二维数组练习2

刷新

#	用户名	姓名	编程分	时间	A	B	C	D	E
1	chuxingyi	初星熠	500	47	100	100	100	100	100
2	liuzichen123	刘子郴	500	47	100	100	100	100	100
3	zhanggao	张高	500	47	100	100	100	100	100
4	yuhaohua	于昊华	500	48	100	100	100	100	100
5	hepeiyu	何佩瑜	500	48	100	100	100	100	100
6	chenzhiheng	陈之恒	500	51	100	100	100	100	100
7	suihaochen	隋溟琛	500	54	100	100	100	100	100
8	huojiaming	霍佳铭	500	60	100	100	100	100	100
9	liuyinan	刘奕男	450	48	100	100	100	100	50
10	zangchenyun	臧晨云	400	44	100	100	100	100	
11	lijingxuan	李景翾	400	46	100	100	100	100	0
12	liukeyang	刘克阳	400	51	100	100	100	100	0

作业

https://cppoj.kids123code.com/contest/4162 (课上讲了 A ~ H 题, 作业是 A ~ H 题必做)

课堂表现

今天上课给同学们讲解了 char 和 string 的简单应用, 同学们课上整体吸收还不错, 课下要加强复习。

课堂内容

判断是什么字符

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
  char c;
  cin >> c;
  if (c>='A' && c<='Z') {
    cout << "upper";
```

```
  }
  else if (c>='a' && c<='z') {
    cout << "lower";
  }
  else {
    cout << "digit";
  }
  return 0;
}
```

大小写转换

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
  char c;
  cin >> c;
  if (c>='A' && c<='Z') {
    cout << char(c-'A'+'a');
  }
  else {
    cout << char(c-'a'+'A');
  }
  return 0;
}
```

打印小写字母表

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
  for (char i = 'a'; i <= 'm'; i++) {
    cout << i;
  }
  cout << endl;
  for (char i = 'n'; i <= 'z'; i++) {
    cout << i;
  }
  cout << endl;
  for (char i = 'z'; i >= 'n'; i--) {
    cout << i;
  }
```

```
  cout << endl;
  for (char i = 'm'; i >= 'a'; i--) {
    cout << i;
  }
  return 0;
}
```

石头剪子布

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
  int n;
  cin >> n;
  for (int i = 1; i <= n; i++) {
    string a, b;
    cin >> a >> b;
    if (a == b) {
      cout << "Tie" << endl;
    }
    else if ((a=="Rock"&&b=="Scissors") || (a=="Scissors"&&b=="Paper") ||
(a=="Paper"&&b=="Rock")) {
      cout << "Player1" << endl;
    }
    else {
      cout << "Player2" << endl;
    }
  }
  return 0;
}
```

判断字符串是否为回文

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
  string a, b;
  cin >> a;
  b = a;
  reverse(b.begin(), b.end());
  if (a == b) {
    cout << "yes" << endl;
```

```
  }
  else {
    cout << "no" << endl;
  }
  return 0;
}
```

【深基6.例1】自动修正

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
  string s;
  cin >> s;
  int n = s.size();
  s = " " + s;
  for (int i = 1; i <= n; i++) {
    if (s[i]>='a' && s[i]<='z') {
      s[i] = s[i]-'a'+'A';
    }
  }
  for (int i = 1; i <= n; i++) {
    cout << s[i];
  }
  return 0;
}
```

配对碱基链

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
  string s;
  cin >> s;
  int n = s.size();
  s = " " + s;
  for (int i = 1; i <= n; i++) {
    if (s[i] == 'A') {
      cout << "T";
    }
    else if (s[i] == 'T') {
      cout << "A";
```

```
    }
    else if (s[i] == 'G') {
      cout << "C";
    }
    else {
      cout << "G";
    }
  }
  return 0;
}
```

基因相关性

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
  double t;
  cin >> t;

  string a, b;
  cin >> a >> b;
  int n = a.size();
  a = " " + a, b = " " + b;

  int cnt = 0;
  for (int i = 1; i <= n; i++) {
    if (a[i] == b[i]) {
      cnt++;
    }
  }

  double z = 1.0*cnt/n;
  if (z >= t) {
    cout << "yes" << endl;
  }
  else {
    cout << "no" << endl;
  }
  return 0;
}
```